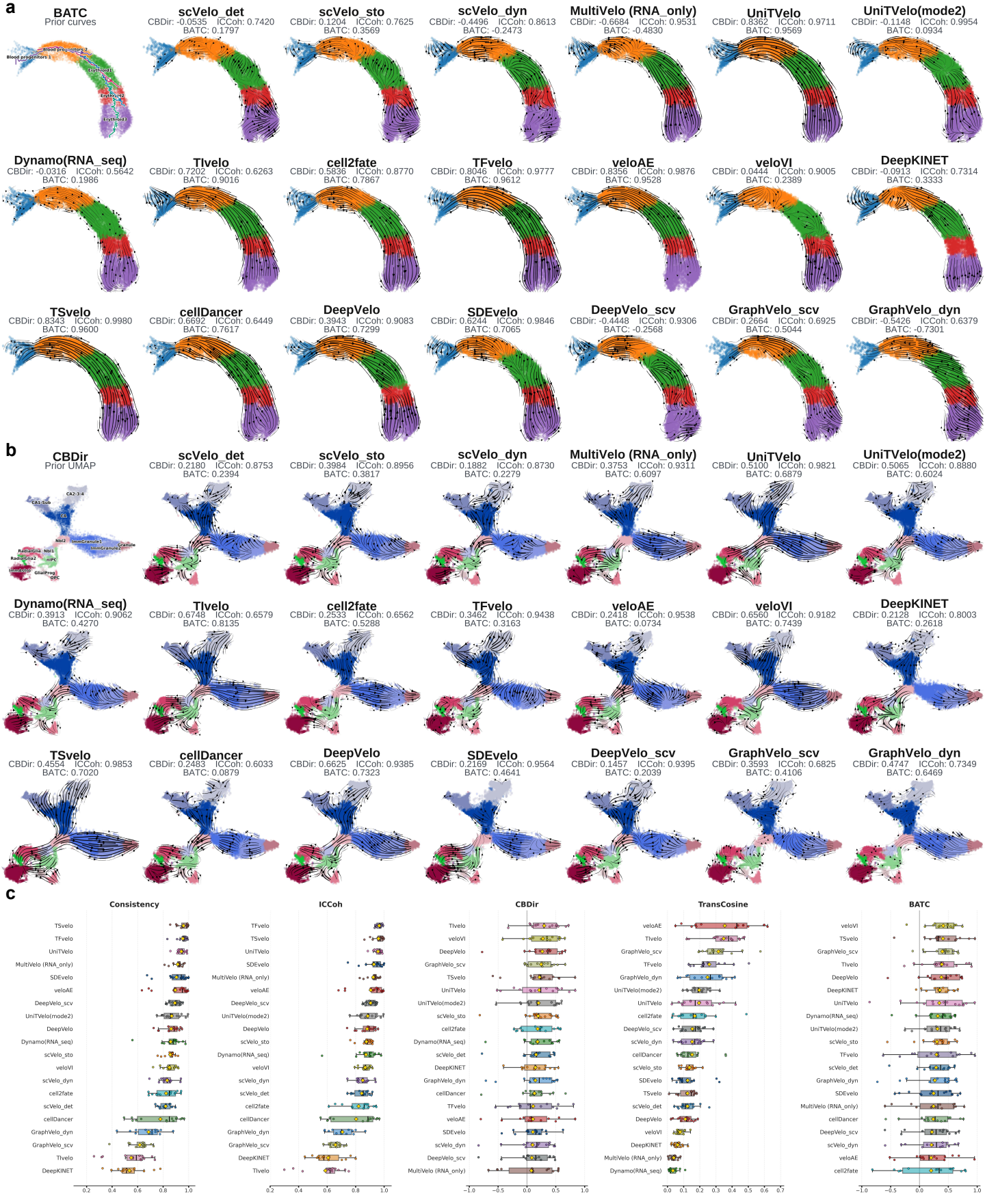

a
BATC
Prior curves
Blood Prog_progen 2
Blood progenitors 1
Erythroid1
Erythroid2
Erythroid3
scVelo_det
CBDir: -0.0535 ICCoh: 0.7420
BATC: 0.1797
scVelo_sto
CBDir: 0.1204 ICCoh: 0.7625
BATC: 0.3569
scVelo_dyn
CBDir: -0.4496 ICCoh: 0.8613
BATC: -0.2473
MultiVelo (RNA_only)
CBDir: -0.6684 ICCoh: 0.9531
BATC: -0.4830
UniTVelo
CBDir: 0.8362 ICCoh: 0.9711
BATC: 0.9569
UniTVelo(mode2)
CBDir: -0.1148 ICCoh: 0.9954
BATC: 0.0934
Dynamo(RNA_seq)
CBDir: -0.0316 ICCoh: 0.5642
BATC: 0.1986
TIvelo
CBDir: 0.7202 ICCoh: 0.6263
BATC: 0.9016
cell2fate
CBDir: 0.5836 ICCoh: 0.8770
BATC: 0.7867
TFvelo
CBDir: 0.8046 ICCoh: 0.9777
BATC: 0.9612
veloAE
CBDir: 0.8356 ICCoh: 0.9876
BATC: 0.9528
veloVI
CBDir: 0.0444 ICCoh: 0.9005
BATC: 0.2389
DeepKINET
CBDir: -0.0913 ICCoh: 0.7314
BATC: 0.3333
TSvelo
CBDir: 0.8343 ICCoh: 0.9980
BATC: 0.9600
cellDancer
CBDir: 0.6692 ICCoh: 0.6449
BATC: 0.7617
DeepVelo
CBDir: 0.3943 ICCoh: 0.9083
BATC: 0.7299
SDEvelo
CBDir: 0.6244 ICCoh: 0.9846
BATC: 0.7065
DeepVelo_scv
CBDir: -0.4448 ICCoh: 0.9306
BATC: -0.2568
GraphVelo_scv
CBDir: 0.2664 ICCoh: 0.6925
BATC: 0.5044
GraphVelo_dyn
CBDir: -0.5426 ICCoh: 0.6379
BATC: -0.7301
b
CBDir
Prior UMAP
CA2-3-4
CA1-Sub
DG
Nbl2
ImmGranule1
Granule
ImmGranule2
Radial-Glia
Nbl1
Radial-Glia2
nIPC
ImmAstro
GlialProg
OPC
scVelo_det
CBDir: 0.2180 ICCoh: 0.8753
BATC: 0.2394
scVelo_sto
CBDir: 0.3984 ICCoh: 0.8956
BATC: 0.3817
scVelo_dyn
CBDir: 0.1882 ICCoh: 0.8730
BATC: 0.2279
MultiVelo (RNA_only)
CBDir: 0.3753 ICCoh: 0.9311
BATC: 0.6097
UniTVelo
CBDir: 0.5100 ICCoh: 0.9821
BATC: 0.6879
UniTVelo(mode2)
CBDir: 0.5065 ICCoh: 0.8880
BATC: 0.6024
Dynamo(RNA_seq)
CBDir: 0.3913 ICCoh: 0.9062
BATC: 0.4270
TIvelo
CBDir: 0.6748 ICCoh: 0.6579
BATC: 0.8135
cell2fate
CBDir: 0.2533 ICCoh: 0.6562
BATC: 0.5288
TFvelo
CBDir: 0.3462 ICCoh: 0.9438
BATC: 0.3163
veloAE
CBDir: 0.2418 ICCoh: 0.9538
BATC: 0.0734
veloVI
CBDir: 0.6560 ICCoh: 0.9182
BATC: 0.7439
DeepKINET
CBDir: 0.2128 ICCoh: 0.8003
BATC: 0.2618
TSvelo
CBDir: 0.4554 ICCoh: 0.9853
BATC: 0.7020
cellDancer
CBDir: 0.2483 ICCoh: 0.6033
BATC: 0.0879
DeepVelo
CBDir: 0.6625 ICCoh: 0.9385
BATC: 0.7323
SDEvelo
CBDir: 0.2169 ICCoh: 0.9564
BATC: 0.4641
DeepVelo_scv
CBDir: 0.1457 ICCoh: 0.9395
BATC: 0.2039
GraphVelo_scv
CBDir: 0.3593 ICCoh: 0.6825
BATC: 0.4106
GraphVelo_dyn
CBDir: 0.4747 ICCoh: 0.7349
BATC: 0.6469
c
Consistency
TSvelo
TFvelo
UniTVelo
MultiVelo (RNA_only)
SDEvelo
veloAE
DeepVelo_scv
UniTVelo(mode2)
DeepVelo
Dynamo(RNA_seq)
scVelo_sto
veloVI
scVelo_dyn
cell2fate
scVelo_det
cellDancer
GraphVelo_dyn
GraphVelo_scv
TIvelo
DeepKINET
0.2
0.4
0.6
0.8
1.0
ICCoh
TFvelo
TSvelo
UniTVelo
SDEvelo
MultiVelo (RNA_only)
veloAE
DeepVelo_scv
UniTVelo(mode2)
DeepVelo
scVelo_sto
Dynamo(RNA_seq)
veloVI
scVelo_dyn
scVelo_det
cell2fate
cellDancer
GraphVelo_dyn
GraphVelo_scv
DeepKINET
TIvelo
0.2
0.4
0.6
0.8
1.0
CBDir
TIvelo
veloVI
DeepVelo
GraphVelo_scv
TSvelo
UniTVelo
UniTVelo(mode2)
scVelo_sto
cell2fate
Dynamo(RNA_seq)
scVelo_det
DeepKINET
GraphVelo_dyn
cellDancer
TFvelo
veloAE
SDEvelo
scVelo_dyn
DeepVelo_scv
MultiVelo (RNA_only)
−1.0
−0.5
0.0
0.5
1.0
TransCosine
veloAE
TIvelo
GraphVelo_scv
TFvelo
GraphVelo_dyn
UniTVelo(mode2)
UniTVelo
cell2fate
DeepVelo_scv
scVelo_dyn
cellDancer
scVelo_sto
SDEvelo
TSvelo
scVelo_det
DeepVelo
veloVI
DeepKINET
MultiVelo (RNA_only)
Dynamo(RNA_seq)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
BATC
veloVI
TSvelo
GraphVelo_scv
TIvelo
DeepVelo
DeepKINET
UniTVelo
Dynamo(RNA_seq)
UniTVelo(mode2)
scVelo_sto
TFvelo
scVelo_det
GraphVelo_dyn
SDEvelo
MultiVelo (RNA_only)
cellDancer
DeepVelo_scv
scVelo_dyn
veloAE
cell2fate
−1.0
−0.5
0.0
0.5
1.0